

A lower bound for 1-median selection in metric spaces

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Abstract

Consider the problem of finding a point in an n -point metric space with the minimum average distance to all points. We show that this problem has no deterministic nonadaptive $o(n^2)$ -query $(4 - \Omega(1))$ -approximation algorithms.

erage distance to other actors [13]. So METRIC 1-MEDIAN can be interpreted as the problem of finding the most important point in a metric space. Goldreich and Ron [6] and Eppstein and Wang [5] present randomized algorithms for approximating the closeness centralities of vertices in undirected graphs.

1 Introduction

Given oracle access to a metric space $(\{1, 2, \dots, n\}, d)$, the METRIC 1-MEDIAN problem asks for a point with the minimum average distance to all points. Indyk [8, 9] shows that METRIC 1-MEDIAN has a Monte-Carlo $O(n/\epsilon^2)$ -time $(1 + \epsilon)$ -approximation algorithm with an $\Omega(1)$ probability of success. The more general METRIC k -MEDIAN problem asks for $x_1, x_2, \dots, x_k \in \{1, 2, \dots, n\}$ minimizing $\sum_{x \in \{1, 2, \dots, n\}} \min_{i=1}^k d(x_i, x)$. Randomized as well as evasive algorithms are well-studied for METRIC k -MEDIAN and the related k -means problem [1, 4, 7, 10–12], where $k \geq 1$ is part of the input rather than a constant.

This paper focuses on *deterministic sublinear-query* algorithms for METRIC 1-MEDIAN. Guha et al. [7, Sec. 3.1–3.2] prove that METRIC k -MEDIAN has a deterministic $O(n^{1+\epsilon})$ -time $O(n^\epsilon)$ -space $2^{O(1/\epsilon)}$ -approximation algorithm that reads distances in a single pass, where $\epsilon > 0$. Chang [3] presents a deterministic nonadaptive $O(n^{1.5})$ -time 4-approximation algorithm for METRIC 1-MEDIAN. He also shows that METRIC 1-MEDIAN has no deterministic $o(n^2)$ -query $(3 - \epsilon)$ -approximation algorithms for any constant $\epsilon > 0$ [2]. Modifying a proof of Chang [2, Sec. 5], this paper shows that METRIC 1-MEDIAN has no deterministic nonadaptive $o(n^2)$ -query $(4 - \epsilon)$ -approximation algorithms for any constant $\epsilon > 0$.

In social network analysis, the importance of an actor in a network may be quantified by several centrality measures, among which the closeness centrality of an actor is defined to be its av-

2 Definitions

For $n \in \mathbb{Z}^+$, denote $[n] \equiv \{1, 2, \dots, n\}$. An n -point metric space $([n], d)$ is the set $[n]$ endowed with a function $d: [n] \times [n] \rightarrow \mathbb{R}$ satisfying

1. $d(x, y) \geq 0$ (non-negativeness),
2. $d(x, y) = 0$ if and only if $x = y$ (identity of indiscernibles),
3. $d(x, y) = d(y, x)$ (symmetry), and
4. $d(x, y) + d(x, z) \geq d(y, z)$ (triangle inequality)

for all $x, y, z \in [n]$. An equivalent definition requires the triangle inequality only for distinct $x, y, z \in [n]$, axioms 1–3 remaining.

An algorithm with oracle access to a metric space $([n], d)$ is given n and may query d on any $(x, y) \in [n] \times [n]$ to obtain $d(x, y)$. Without loss of generality, we forbid queries for $d(x, x)$, which trivially returns 0, as well as repeated queries, where queries for $d(x, y)$ and $d(y, x)$ are seen as identical. An algorithm is said to be nonadaptive if its $(k + 1)$ -th query (which is a pair in $[n] \times [n]$) is independent of the answers to the first k queries, where $k \in \mathbb{N}$. Consequently, a deterministic nonadaptive algorithm's set of queries depends only on n and the algorithm itself but not on d . For convenience, denote an algorithm ALG with oracle access to $([n], d)$ by ALG^d .

Given oracle access to a finite metric space $([n], d)$, the METRIC 1-MEDIAN problem asks for a point in $[n]$ with the minimum average distance

to all points. An algorithm for this problem is α -approximate if it outputs a point $x \in [n]$ satisfying

$$\sum_{y \in [n]} d(x, y) \leq \alpha \min_{x' \in [n]} \sum_{y \in [n]} d(x', y),$$

where $\alpha \geq 1$.

The following theorem is due to Chang [3].

Theorem 1 ([3]). *METRIC 1-MEDIAN has a deterministic nonadaptive $O(n^{1.5})$ -time 4-approximation algorithm.*

3 Lower bound

Let A be any deterministic nonadaptive $o(n^2)$ -query algorithm for METRIC 1-MEDIAN, $\epsilon \in (0, 0.1)$ be a constant and $d: [n] \times [n] \rightarrow \mathbb{R}$ be a metric to be determined later.

Define

$$Q \equiv \left\{ \text{unordered pair } (x, y) \in [n]^2 \mid A^d \text{ ever queries for } d(x, y) \right\}$$

to be the set of queries of A^d treated as unordered pairs. Without loss of generality, assume $(x, x) \notin Q$ for all $x \in [n]$. Let $G = ([n], Q)$ be the simple undirected graph with vertex set $[n]$ and edge set Q . Denote the degree of $x \in [n]$ in G by $\deg_G(x) = |\{y \in [n] \mid (x, y) \in Q\}|$, and

$$B \equiv \{x \in [n] \mid \deg_G(x) \geq \epsilon n\}. \quad (1)$$

In the sequel, we will specify d incrementally in several steps. Note that Q and B are independent of d because of the nonadaptivity of A ; hence they will remain intact during our specification of d .

Below is an easy lemma.

Lemma 2 (Implicit in [2]). $|B| = o(n)$.

Henceforth we will assume $n \in \mathbb{Z}^+$ to be sufficiently large so that

$$n - |B| - 1 - \epsilon n > 0 \quad (2)$$

by Lemma 2. For all $x \in [n]$,

$$d(x, x) \equiv 0. \quad (3)$$

For all $(x, y) \in [n]^2 \setminus \{(x, x) \mid x \in [n]\}$ with $x \in B$, $y \in B$ or $(x, y) \in Q$,

$$d(x, y) \equiv \begin{cases} 4, & \text{if } x \in B \text{ or } y \in B; \\ 2, & \text{otherwise.} \end{cases} \quad (4)$$

Clearly, this does not assign different values to $d(x, y)$ and $d(y, x)$.

As Eq. (4) specifies d on a superset of Q (which is the set of A 's queries) and A is deterministic, the output of A^d has now been fixed even though d is not fully specified yet. Let $p \in [n] \setminus B$ and $p' \in B$ be such that $\{p, p'\}$ contains the output of A^d .

Lemma 3.

$$|([n] \setminus (B \cup \{p\})) \cap \{x \in [n] \mid (p, x) \notin Q\}| \geq n - |B \cup \{p\}| - \epsilon n.$$

Proof. Eq. (1) and $p \notin B$ imply $\deg_G(p) < \epsilon n$, i.e., $|\{x \in [n] \mid (p, x) \in Q\}| < \epsilon n$. $\square$

Take

$$\hat{p} \in ([n] \setminus (B \cup \{p\})) \cap \{x \in [n] \mid (x, p) \notin Q\} \quad (5)$$

arbitrarily, as can be done by Lemma 3 and Eq. (2). Trivially, $\hat{p} \notin B$.

We now complete the specification of d . For all $(x, y) \in [n]^2 \setminus (Q \cup \{(x, x) \mid x \in [n]\})$ with $x \notin B$ and $y \notin B$,¹

$$d(x, y) \equiv \begin{cases} 3, & \text{if } ((x = \hat{p}) \wedge (y = p)) \text{ or } ((y = \hat{p}) \wedge (x = p)); \\ 1, & \text{if } ((x = \hat{p}) \wedge (y \neq p)) \text{ or } ((y = \hat{p}) \wedge (x \neq p)); \\ 4, & \text{if } ((x = p) \wedge (y \neq \hat{p})) \text{ or } ((y = p) \wedge (x \neq \hat{p})); \\ 2, & \text{otherwise.} \end{cases} \quad (6)$$

The four cases in Eq. (6) are mutually exclusive because $p \neq \hat{p}$ by Eq. (5). Clearly, Eq. (6) does not assign different values to $d(x, y)$ and $d(y, x)$.

The following lemma is straightforward from Eqs. (3)–(4) and (6).

Lemma 4. For all $x, y \in [n]$, $d(x, y) \in \{0, 1, 2, 3, 4\}$.

Lemma 5.

$$\sum_{y \in [n]} d(\hat{p}, y) \leq (1 + 4\epsilon)n + o(n).$$

Proof. By Lemmas 2 and 4,

$$\sum_{y \in B} d(\hat{p}, y) = o(n). \quad (7)$$

Furthermore,

$$\begin{aligned} & \sum_{y \in [n] \text{ s.t. } (\hat{p}, y) \in Q} d(\hat{p}, y) \\ & \stackrel{\text{Lemma 4}}{\leq} \sum_{y \in [n] \text{ s.t. } (\hat{p}, y) \in Q} 4 \\ & = 4 \deg_G(\hat{p}) \\ & < 4\epsilon n, \end{aligned} \quad (8)$$

¹These are precisely the pairs whose d -distances are not specified by Eqs. (3)–(4).

where the last inequality follows from Eq. (1) and $\hat{p} \notin B$. We have

$$\sum_{y \in [n] \setminus (B \cup \{p, \hat{p}\}) \text{ s.t. } (\hat{p}, y) \notin Q} d(\hat{p}, y) \leq n \quad (9)$$

because all summands are 1 by Eq. (6) and $\hat{p} \notin B$. By Lemma 4,

$$\sum_{y \in \{p, \hat{p}\}} d(\hat{p}, y) = O(1). \quad (10)$$

Summing up Eqs. (7)–(10) completes the proof. $\square$

Lemma 6.

$$\sum_{y \in [n]} d(p, y) \geq 4(n - o(n) - \epsilon n).$$

Proof. Recall that $p \notin B$. We have

$$\begin{aligned} & \sum_{y \in [n]} d(p, y) \\ & \geq \sum_{y \in [n] \setminus (B \cup \{p, \hat{p}\}) \text{ s.t. } (p, y) \notin Q} d(p, y) \\ & \stackrel{\text{Eq. (6)}}{=} \sum_{y \in [n] \setminus (B \cup \{p, \hat{p}\}) \text{ s.t. } (p, y) \notin Q} 4 \\ & \geq 4(|\{y \in [n] \setminus (B \cup \{p\}) \mid (p, y) \notin Q\}| - 1) \\ & \stackrel{\text{Lemma 3}}{\geq} 4(n - |B| - \epsilon n - 2) \\ & \stackrel{\text{Lemma 2}}{=} 4(n - o(n) - \epsilon n). \end{aligned}$$

$\square$

The next lemma is immediate from Eq. (4) and $p' \in B$.

Lemma 7.

$$\sum_{y \in [n] \setminus \{p'\}} d(p', y) = 4(n - 1).$$

We proceed to prove that $([n], d)$ is a metric space through a few lemmas.

Lemma 8. *For all $x, y \in [n]$, if $d(x, y) = 1$, then $\hat{p} \in \{x, y\}$.*

Proof. Inspect Eq. (6), which is the only equation that may set distances to 1. $\square$

Below is a consequence of Lemma 8.

Lemma 9. *For all distinct $x, y, z \in [n]$, if $d(x, y) = d(x, z) = 1$, then $x = \hat{p}$.*

The following lemma is immediate from Eqs. (4) and (6).

Lemma 10. *For all $x, y \in [n]$, if $d(x, y) = 4$, then $\{x, y\} \cap (B \cup \{p\}) \neq \emptyset$.*

Below is a consequence of $p \notin B$ and Eqs. (5)–(6).

Lemma 11. $d(\hat{p}, p) = 3$.

Lemma 12. $([n], d)$ is a metric space.

Proof. We only need to prove the triangle inequality for d because all the other axioms are easy to verify. Consider the following cases for all distinct $x, y, z \in [n]$:

- $d(x, y) = 1, d(x, z) = 1$ and $d(y, z) = 4$. By Lemma 9, $x = \hat{p}$. Hence if $y = p$ (resp., $z = p$), then $d(x, y) = 3$ (resp., $d(x, z) = 3$) by Lemma 11, a contradiction. Therefore, $p \notin \{y, z\}$, which together with Lemma 10 forces $\{y, z\} \cap B \neq \emptyset$. But if $y \in B$ (resp., $z \in B$), then $d(x, y) = 4$ (resp., $d(x, z) = 4$) by Eq. (4), a contradiction.
- $d(x, y) = 1, d(x, z) = 1$ and $d(y, z) = 3$. By Lemma 9, $x = \hat{p}$. On the other hand, $d(y, z) = 3$ means $(y, z) \in \{(\hat{p}, p), (p, \hat{p})\}$ by Eq. (6) (which is the only equation that may set distances to 3), contradicting $x = \hat{p}$.
- $d(x, y) = 1, d(x, z) = 2$ and $d(y, z) = 4$. By Lemma 10, $\{y, z\} \cap (B \cup \{p\}) \neq \emptyset$. But if $y \in B$ (resp., $z \in B$), then $d(x, y) = 4$ (resp., $d(x, z) = 4$) by Eq. (4), a contradiction. Therefore, $p \in \{y, z\}$. Furthermore, $\hat{p} \in \{x, y\}$ by Lemma 8. Consequently, $(p, \hat{p}) \in \{(x, y), (x, z), (y, z)\}$ (note that $p \neq \hat{p}$ by Eq. (5)), implying $3 \in \{d(x, y), d(x, z), d(y, z)\}$ by Lemma 11, a contradiction.

We have excluded all possibilities of $d(x, y) + d(x, z) < d(y, z)$, where $x, y, z \in [n]$. $\square$

Combining Lemmas 5–7, 12 and that $\{p, p'\}$ contains the output of A^d yields our main theorem.

Theorem 13. METRIC 1-MEDIAN has no deterministic nonadaptive $o(n^2)$ -query $(4 - \epsilon)$ -approximation algorithms for any constant $\epsilon > 0$.

Theorem 13 shows that the approximation ratio of 4 in Theorem 1 cannot be improved to any constant $c < 4$.

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